Step 2b - H5 - Microwave access simulation

Tech baseline

Technologies used in the market

Expert-level generalist EM/multiphysics tools (Ansys HFSS, CST, COMSOL RF)

How it works: The theoretically-correct path: model applicator + product + thermal coupling; in practice requires hiring or contracting EM specialists plus $10-70k seats.

How well it performs: Validated accuracy; complete design freedom; combined license+expert cost easily ~$150-300k first year for a new OEM

Limits: This is the barrier, not the solution: steep learning curve, no microwave-heating defaults, and no guidance on dielectric data or ISM compliance — unusable for a first-time OEM without external help

Template/app-based simulation (COMSOL Application Builder + compiled apps, QuickWave BHM projects)

How it works: Experts encapsulate validated models behind simplified UIs (product dimensions, moisture, throughput → field/uniformity results); COMSOL Server/Compiler can distribute apps to non-specialists.

How well it performs: Proven pattern (ITW built internal MW-heating apps on COMSOL; frozen-sandwich simulator case study); reduces user skill need while keeping full-wave accuracy

Limits: Requires an expert to build each app (service-heavy); per-domain templates for industrial applicators do not exist off-the-shelf; licensing costs for app distribution add up

Cloud/browser simulation platforms (SimScale, Ansys Gateway/AWS, COMSOL Server)

How it works: Browser access to solvers on elastic HPC; removes workstation and license-infrastructure barriers.

How well it performs: On-demand compute at ~$ per run vs ~$50k seats; market analysts cite cloud delivery as a key growth driver

Limits: Solves the compute-access problem only; user still needs EM model-building expertise and material data — the actual bottleneck for non-expert OEMs

Component/vendor-led integration (Muegge, SAIREM, Richardson, e2v, CPI generators + applicators + application labs)

How it works: OEM buys standard generators, waveguides and applicators and relies on the component vendor's engineering support and test labs for integration.

How well it performs: The dominant real-world path today; vendors advertise application support across sectors; SAIREM alone cites ~5,000 machines installed

Limits: Locks the OEM into one supplier's catalogue and margins; limited physics insight into the OEM's specific product/load; vendors have no incentive to optimize outside their component envelope

Consultants, university labs and feasibility testing (AMPERE/IMPI networks)

How it works: Contract research: dielectric measurement, pilot trials on vendor/university equipment, bespoke applicator design.

How well it performs: Real expertise on demand; typical feasibility projects ~$20-100k and months of calendar time

Limits: Not scalable, knowledge does not accumulate in the OEM; each new application restarts the same learning curve

Open-source EM codes (openEMS, MEEP, gprMax)

How it works: Free FDTD/FEM solvers, script-driven, usable as the engine inside a guided product.

How well it performs: Zero license cost; adequate physics for 2.45 GHz cavities; growing validation literature

Limits: Expert-only interfaces, no heating/materials workflow, no support — not viable for target customers directly, but a viable compute backend for a ProSeed product

Why our solution can be better

The barrier for non-microwave OEMs is not solver access — it is the missing expertise and the missing dielectric/compliance data. ProSeed can beat incumbents by packaging what the component vendors and consultants sell implicitly (validated applicator templates, measured dielectric database, ISM/IEC-60519-6-aware design rules) into a guided tool-plus-service product: feasibility answer in days instead of a $50k study, then a designed applicator the OEM integrates with any generator supplier. Because ProSeed owns real microwave equipment and process data, its templates start calibrated rather than generic — a structural advantage over both cloud CAE platforms and generalist EM suites for this novice segment.

Key assumptions

• The motivated buyer subset is ~300-1,000 OEMs globally, i.e., a small fraction of the ~10k+ thermal-equipment builders — reachable but education-intensive (assumed; consistent with CalNEXT's 'knowledge and awareness' barrier finding)

• First-time microwave integration today costs ~$150-300k in tools+expertise before prototyping, making a sub-$50k guided product attractive (assumed from license costs ~$10-70k + specialist salaries/consulting rates)

• Adjacent thermal-equipment markets (~$5-8B dryers, ~$3B drying ovens) quantify genuine integration headroom vs the ~$1.4-1.6B microwave market today (measured)

• Electrification/decarbonization policy (DOE Industrial Heat Shot names microwave systems; EU microwave reactor projects) creates pull-through demand for OEM product diversification (measured)

• Revenue model must be service-heavy initially (feasibility studies feeding templates); pure self-serve software is unlikely before a template+data library matures (assumed)

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
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Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis